

Research Evidence Package

Project: Cancelable Multimodal Biometric Authentication for Critical Infrastructure **Audited state:** `main @ 873c13e` ("Implement multimodal biometric updates"), 2026-09-24 **Purpose:** the verified facts, numbers, equations and limitations the capstone paper may use, each with its source. This is not the paper.

Every number below carries one of these evidence labels. Never mix them in the paper.

Label	Meaning
[A] REAL-DATA RESULT	Measured on real biometric data (public datasets or real captures)
[B] SYNTHETIC CALIBRATION / SIMULATION	Measured on synthetic vectors through the real code
[C] SOFTWARE TEST	Software correctness evidence, not biometric performance
[D] CONFIGURATION	A value set in code or config, not measured
[E] DERIVATION	Follows mathematically from [B]/[D]
[F] FUTURE	Proposed, not done

Three distinctions that must survive into the paper: 559 passing backend tests $\neq$ 559 authentication trials. The synthetic BioHash calibration curve $\neq$ real-user biometric performance. The face 0.80 / voice 0.75 thresholds are **teacher-requested configuration values**, not experimentally optimized.

1. Executive research summary

- What exists.** A working web system (FastAPI backend + React frontend) that enrolls face and voice (and optionally fingerprint), transforms each embedding into a **256-bit keyed BioHash template**, stores **only** those templates, and authenticates by **Hamming comparison** of templates. Per-modality decisions are expressed as a calibrated **estimate** of face cosine similarity (≥ 0.80) and voice Euclidean distance (≤ 0.75). Decisions are fused under an `ALL_REQUIRED` policy. Templates are revocable through a pool of 4 template sets. Users register with a display name that is kept separate from the internal ID.
- Strongest legitimate numeric evidence.**
 - [A]** Voice embedding model on a held-out VoxCeleb1-subset split (raw embeddings, not templates): EER 2.29 %, AUC 0.9967 over 281,625 pairs.
 - [B]** The Hamming $\rightarrow$ cosine calibration fits with $R^2 \geq 0.99986$ and a cosine-space RMSE of 0.004–0.005.
 - [B]** Template-set experiments on synthetic embeddings: cross-key similarity 0.500 ± 0.029 (unlinkable-looking), revocation drops the old template to 0.504, 80/80 promotions accepted.
 - [C]** 559/559 backend tests pass.
- What does not exist.** No labelled genuine/impostor evaluation of the **deployed protected-template system** on real users. So **FAR, FRR, EER and ROC-AUC of the actual system cannot be established from the current repository**. There is also no formal security evaluation and no latency benchmark. The only timing data is incidental audit-log latency.
- Paper readiness:** suitable today for a **system-design / implementation paper** with honest limitations. **Not** suitable for any claim of authentication accuracy of the deployed system (see §18).

2. Repository architecture (as implemented)

```
Browser (React, frontend/) --multipart--> FastAPI (backend/main.py)
Register: POST /users -> /enroll/face (5 poses) -> /enroll (voice x2)
Verify:   POST /authenticate/fusion (any enrolled subset)
      |
      | backend/services/authentication.py::authenticate_samples
      |   | per modality
      | backend/services/base_service.py::ModalityService
      |   embed (embeddings/pipelines.py) -> BioHash (template_protection/biohash.py)
      |   -> Hamming compare (template_protection/matcher.py)
      |   -> decision (backend/services/modality_metrics.py)
      |
      | fusion/policy.py::evaluate_fusion_policy (ALL_REQUIRED default)
      |
      | SQLite (backend/database/models.py): users, protected_templates, audit_logs, enrollment_events
```

2.1 Face pipeline

Stage	File / function	Input → output	Parameters	Label
Capture	frontend/src/components/capture/GuiFaceCapture.tsx	webcam frame → PNG	5 poses: front, left, right, up, down	[D]
Detection + crop	preprocessing/face.py::FacePreprocessor (MTCNN, facenet-pytorch)	RGB → 160×160 crop	image_size=160, margin=0, post_process=True	[D]
Enrollment quality gates	preprocessing/face.py, embeddings/pipelines.py	crop → VALID/BLURRY/...	Laplacian sharpness ≥ 25.0; detection prob ≥ 0.90; face-size ratio ≥ 0.15; centre offset ≤ 0.35; roll ≤ 20°; yaw ratio ≤ 0.20; ≥ 3 valid of 5 (MIN_VALID_POSES)	[D]
Embedding	models/face/inference.py::FaceEmbedder (InceptionResnetV1)	160×160×3 → $\mathbb{R}^{512}$	checkpoint models/face/saved/face_embedder.pt (Git LFS)	[D]
Normalization	models/common/base_embedder.py::_l2_normalize	→ unit vector	$\ e\ _2 = 1$	[D]
Enrollment centroid	embeddings/centroid.py::centroid_embedding	valid pose embeddings → one unit vector	normalize(mean(e _i))	[D]
BioHash	template_protection/biohash.py::generate_template	$\mathbb{R}^{512} \rightarrow \{0,1\}^{256}$	see §2.3	[D]
Compare + decide	matcher.compare, modality_metrics.decide	two templates → $\hat{c}$, match	$\hat{c} \geq 0.80$	[D]

Model provenance (documented, not re-verifiable from committed artefacts): InceptionResnetV1 pretrained on VGGFace2 and fine-tuned with an ArcFace head on LFW (62 identities, 3,023 images, min_faces_per_person=20) (docs/PROJECT_REPORT.md §8.2).

2.2 Voice pipeline

Stage	File / function	Input → output	Parameters	Label
Capture	frontend/.../VoiceCapture.tsx, useWavRecorder.ts	mic → WAV	4–5 s, auto-stop at 5 s; phrase "Security authentication for government access."	[D]
Preprocessing	preprocessing/voice.py	waveform → 80 × frames log-mel	16 kHz resample, mono, energy VAD (ratio 0.02), RMS normalization (target 0.1), 4.0 s segment, n_fft 400, hop 160, 80 mel	[D]
Embedding	models/voice/inference.py::VoiceEmbedder (ECAPA-TDNN, models/voice/model.py)	log-mel → $\mathbb{R}^{192}$	checkpoint voice_embedder.pt; trained with ArcFace m = 0.5, s = 30 (models/voice/saved/training_config.json)	[D]
Normalization	base_embedder._l2_normalize	→ unit vector	$\ e\ _2 = 1$ (verified in code; this is what makes $d = \sqrt{2-2\cos}$ valid)	[D]
Enrollment consistency	backend/services/recording_quality.py	two recordings → band	exact cosine of two in-memory embeddings: EXCELLENT ≥ 0.85, GOOD ≥ 0.75, FAIR ≥ 0.60 (needs confirmation), POOR < 0.60 (rejected); the template is built from recording 1	[D]
BioHash	same as face	$\mathbb{R}^{192} \rightarrow \{0,1\}^{256}$	256 > 192, so the projection uses two orthonormal blocks (192 + 64 rows), not mutually orthogonal (transform.build_orthonormal_projection)	[D]
Compare + decide	modality_metrics.decide	→ $\hat{d}$, match	$\hat{d} \leq 0.75$	[D]

2.3 Cancelable transform (BioHash) — template_protection/

- Key material** (hkdf_keys.py::derive_key): salt = SHA-256(application_id|user_id|modality|key_version); three HKDF-SHA256 expansions of MASTER_SECRET with distinct info labels → 32-byte projection_seed, permutation_seed, threshold_seed.
- Normalize:** $x \leftarrow x/\|x\|$.
- Project** (transform.py): $W \in \mathbb{R}^{\wedge\{256 \times D\}}$, rows orthonormal within blocks of ≤ D (QR of a seeded Gaussian, i.e. Haar-random); $p = Wx$, each $p_i \in [-1, 1]$.
- Quantize:** $b_i = 1[p_i > t_i]$, $t_i \sim N(0, (0.5 \cdot \text{std}(p))^2)$, seeded by threshold_seed.
- Permute:** a keyed permutation π of the 256 bits.
- Store:** pack_bits, 256 bits = **32 bytes** (verified: length(protected_template)=32 for every row in the local DB).

TEMPLATE_FORMAT_VERSION = 1. The library default is DEFAULT_OUTPUT_BITS = 128, but runtime always passes Settings.template_bits = 256; the stored row keeps its own output_bits.

2.4 Comparison, decision and fusion

- `matcher.compare(a, b, "hamming")` = fraction of equal bits (1.0 on constant-time exact equality).
- `modality_metrics.decide` converts that into the modality's metric and decision (§8).
- `authentication.authenticate_samples`: every submitted modality is authenticated against the **ACTIVE** template set (never STANDBY/REVOKED, and never sets mixed), then fused by `fusion/policy.py` (§8).

3. Complete numerical evidence sheet (traceability)

Format: **name** → **value** → **source** → **interpretation**.

3.1 Configuration [D]

- Face embedding dimension → 512 → `models/face/inference.py::FACE_EMBEDDING_DIM` → BioHash input size
- Voice embedding dimension → 192 → `models/voice/inference.py::VOICE_EMBEDDING_DIM` → BioHash input size
- Fingerprint embedding dimension → **512** → `models/fingerprint/inference.py::FINGERPRINT_EMBEDDING_DIM` (since `fc9fd47`) → optional modality. **Note:** the calibration script calibrates fingerprint at D = 256 (MODALITY_DIMS), a mismatch (see §15 #8)
- Template length → 256 bits → `backend/config.py::Settings.template_bits` → stored template size
- Stored template size → 32 bytes → local DB `protected_templates.protected_template` → 256/8
- Template sets per user → 4 → `Settings.template_pool_size` → 1 ACTIVE + 3 STANDBY
- Face threshold → 0.80 (estimated cosine, higher = better) → `Settings.face_cosine_threshold` → teacher-requested
- Voice threshold → 0.75 (estimated Euclidean distance, lower = better) → `Settings.voice_euclidean_threshold` → teacher-requested
- Fingerprint threshold → 0.90 (Hamming similarity) → `Settings.match_threshold` fallback (`backend/threshold_loader.py`) → inherited default
- Default fusion policy → ALL_REQUIRED → `fusion/config.py::DEFAULT_FUSION_POLICY`
- Fusion weights → equal (1.0 each, renormalized) → `fusion/score_fusion.py::resolve_normalized_weights`
- WEIGHTED veto floor → 0.0 (estimated-cosine scale) → `fusion/config.py::DEFAULT_WEIGHTED_FLOOR`
- HKDF → SHA-256, 32-byte seeds → `template_protection/hkdf_keys.py`
- Quantization jitter → $\sigma = 0.5 \cdot \text{std}(p)$ → `transform.quantize`
- Face input size → 160×160 → `preprocessing/face.py::FACE_INPUT_SIZE`
- Voice audio → 16 kHz, 4.0 s, 80 mel, `n_fft` 400, hop 160 → `preprocessing/voice.py`
- Upload limit → 5,000,000 bytes → `Settings.max_upload_size_bytes`
- Display name → 1–64 chars; letters, spaces and ' ' - . ; must start with a letter → `backend/display_names.py`

3.2 Derived thresholds [E] (from the calibration curve; computed in this audit with `modality_metrics.decide / MetricCurve`)

- Face $\hat{c} \geq 0.80 \Leftrightarrow$ Hamming similarity $\geq$ **0.81759** $\Leftrightarrow$ differing bits $\leq$ **46** of 256
- Voice $\hat{d} \leq 0.75 \Leftrightarrow \hat{c} \geq 1 - 0.75^2/2 =$ **0.71875** $\Leftrightarrow$ Hamming similarity $\geq$ **0.78229** $\Leftrightarrow$ differing bits $\leq$ **55** of 256
- Fingerprint $h \geq 0.90 \Leftrightarrow$ differing bits $\leq$ **25** $\Leftrightarrow$ fusion-scale $\hat{c} \geq 0.9384$
- Face+voice fusion threshold = $(0.80 + 0.71875)/2 =$ **0.759375** (informational under ALL_REQUIRED)

3.3 Synthetic calibration [B] — `evaluation/results/biohash_metric_calibration.json`

See §4 for the full table.

3.4 Real-dataset model evaluations [A] — raw embeddings, NOT the protected-template system

- **Voice** (`evaluation/results/voice_metrics.csv`, `voice_confusion_matrix.csv`, `voice_roc.csv`; VoxCeleb1 Indian-celebrity subset, Kaggle, held-out split):
 - test utterances: 751; pairs: 281,625 = C(751, 2) (verified); genuine 13,101; impostor 268,524
 - EER 0.022890 at cosine threshold 0.32214; AUC 0.996724 (recomputed here by trapezoid over 4,303 ROC points; matches the CSV's 0.9967238)

- at the EER threshold: TP 12,801, FN 300, FP 6,144, TN 262,380 → accuracy 0.977119, FAR 0.022881, FRR 0.022899, precision 0.675693, recall 0.977101, F1 0.798914 (every value recomputed from the confusion matrix and matching the CSV)
- number of speakers: **NOT AVAILABLE** (printed at kernel run time, not committed)
- **Face** (evaluation/results/face_metrics.csv; LFW, 62 identities, 3,023 images per docs/PROJECT_REPORT.md): accuracy@EER 0.990, EER 0.010, AUC 0.999 (3 decimals, copied from a training-log line). Pair counts, ROC data, split protocol and whether the evaluation identities were disjoint from the training identities: **NOT AVAILABLE**. Treat as indicative only; likely optimistic.
- **Fingerprint** (fingerprint_metrics.csv; SOCOFing): 900 samples, 404,550 pairs = C(900, 2) (verified), EER 0.3077, AUC 0.7644 (recomputed 0.764447). Weak model and outside the face+voice scope.

3.5 Real captures through the deployed system — local audit log (biometric.db, gitignored, NOT in the repo)

These are **unlabelled**: authenticated is the system's decision at the time, not ground truth.

Regime	Period	Scale	Attempts	Internal IDs	Face score stats	Voice score stats
R0: Hamming, threshold 0.90	2026-09-17	Hamming similarity	2	1	n=2: 0.832, 0.902	n=2: 0.805, 0.844
R1: Hamming, threshold 0.80	2026-09-22 → 09-24 08:10 UTC	Hamming similarity	50	6	n=50: min 0.4609, Q1 0.7148, median 0.7871, Q3 0.8633, max 0.9375, mean 0.7635, sd 0.1276	n=15: min 0.5312, Q1 0.7676, median 0.7852, Q3 0.8340, max 0.9102, mean 0.7865, sd 0.0882
R2: new metric, real camera	2026-09-24 15:32, 15:43 UTC	estimated cosine	2	1	0.8921, 0.8858 (both granted)	—
R2: synthetic smoke test	2026-09-24 15:25–15:37 UTC	fusion scale	5	1	—	0.9985 ×3, −0.200 (clamp) ×2; synthetic Chrome microphone, not a person

Decisions in R1: face-only 14 granted / 21 denied; face+voice 3 granted / 12 denied. **Do not infer the number of people from the internal IDs.** One person may hold several IDs, and several people may have used one.

3.6 Documented but not re-verifiable real observations

- docs/AUTHENTICATION_RELIABILITY_REPORT.md: user OPERATOR-UWKAMM, face 0.754 / 0.766 / 0.820 and voice 0.797 / 0.848 / 0.879 (Hamming, asserted genuine); later "fifteen logged genuine face attempts scored 0.738–0.859 (mean 0.784; 5 of 15 ≥ 0.80)". That audit log is **not** in the current database (0 rows for that ID). The deleted face_threshold.json quoted **0.738–0.820**, which disagrees with 0.738–0.859. **Do not cite either range as a verified result.**

3.7 Latency (incidental, from the audit log) — see §13

3.8 Software [C] — see §6

4. Calibration analysis [B] (synthetic, not biometric performance)

Source: scripts/calibrate_biohash_metric_mapping.py → evaluation/results/biohash_metric_calibration.json; runtime template_protection/metric_estimation.py; decisions backend/services/modality_metrics.py; tests tests/test_modality_metrics.py.

Procedure.

- **Input:** pairs of synthetic unit vectors (a, b) with prescribed cosine c, passed through the real BioHash transform. The fast path is asserted bit-identical to generate_template.
- **Grid:** 49 cosines from −0.2 to 1.0 in steps of 0.025.
- **Keys:** 50 keys × 8 pairs per key per grid point = **400 pairs per point = 19,600 pairs per modality**, seed 0, 256 bits.
- **Why synthetic pairs are valid:** the projection is Haar-random and the jitter is scale-relative, so the transform is rotation-invariant. The distribution of Hamming similarity therefore depends only on (c, D, N). This is a **property of the transform**. It is **not** a claim about real embeddings, and it has not been validated on real embedding pairs (see §18, item 2).

Model (fitted by least squares, no intercept by construction; $h(1) = 1$ exactly):

$$1 - h(c) = (p_1\theta + p_2\theta^2 + p_3\theta^3) / \pi, \quad \theta = \arccos c$$

Inversion: $\hat{c} = h^{-1}(\text{observed } h)$, tabulated over 2,001 points and clamped to [−0.2, 1.0].

Statistic	Face (D=512)	Voice (D=192)	Fingerprint (calibrated at D=256; the real model is 512-d, see §15 #8)	How computed
p_1, p_2, p_3	0.881745, 0.027410, -0.021372	0.885522, 0.019056, -0.017816	0.884612, 0.022578, -0.019118	JSON coefficients
Intercept	none (constrained $h(1)=1$)	none	none	model form
Grid points / pairs	49 / 19,600	49 / 19,600	49 / 19,600	JSON
Fit RMSE (Hamming space, fit vs 49 means)	0.00142	0.00119	0.00116	this audit
Fit MAE (Hamming space)	0.00108	0.00088	0.00093	this audit
Max abs residual (Hamming space)	0.00379	0.00306	0.00296	JSON max_fit_residual (matches)
Min abs residual	0.000000 (at $c=1$)	0.000000	0.000000	this audit
R^2 (Hamming space)	0.999865	0.999905	0.999909	this audit
RMSE of mean-curve inversion (cosine space)	0.00455	0.00403	0.00387	this audit
MAE (cosine space)	0.00331	0.00278	0.00297	this audit
Max abs error (cosine space)	0.0142	0.0118	0.0101	this audit
Single-estimate SD at $c = 0.90 / 0.80 / 0.725 / 0.60$	0.032 / 0.050 / 0.063 / 0.077	0.031 / 0.047 / 0.057 / 0.067	0.030 / 0.047 / 0.054 / 0.069	JSON cosine_estimate_std
Largest single-estimate SD (any c)	0.114	0.106	0.101	JSON
h at $c = 0$ (unrelated, same key)	0.5643	0.5673	0.5638	JSON hamming_similarity_mean
h at clamp floor $c = -0.2$	0.5131 (fit)	0.5130	0.5123	model

Interpretation:

- The **model fit** is near-perfect ($R^2 \approx 0.9999$).
- The **per-attempt uncertainty** is what matters for decisions: about ± 0.05 in cosine at the face threshold. That is inherent to 256-bit quantization. It is not a fit error.

Extrapolation risk:

- Observed h below about 0.513 clamps to $\hat{c} = -0.2$ ($\hat{d} = 1.549$). The smoke test's wrong voice ($h = 0.496$) hit this clamp, so the displayed "-0.20" is a floor, not a measurement.
- Under a **shared** key, unrelated embeddings give $h \approx 0.56$, not 0.50, because the quantization thresholds are shared.

Cross-check with independent earlier data [B]: AUTHENTICATION_RELIABILITY_REPORT.md (25 pairs per row) gives $h(0.80) = 0.814$ (face) and 0.816 (voice), against 0.8186 / 0.8172 here. $h(0.00) = 0.547 / 0.558$ against 0.564 / 0.567. The two agree within sampling error.

5. Biometric evaluation analysis

Question	Answer
Labelled genuine/impostor trials of the deployed protected-template system	NOT AVAILABLE FROM CURRENT REPOSITORY
FAR / FRR / EER / ROC-AUC / TAR of the deployed system (face, voice, fusion)	NOT AVAILABLE
Accuracy / precision / recall / F1 of the deployed system	NOT AVAILABLE
Number of real users	NOT AVAILABLE (7 internal IDs in the audit log $\neq$ people)
Enrollment samples per user	By design: face 5 captures (≥ 3 valid) $\rightarrow$ 1 centroid; voice 2 recordings $\rightarrow$ template from recording 1 [D]
Real attempts (unlabelled)	54 non-synthetic audit rows ($R0\ 2 + R1\ 50 + R2\ 2$): 37 face-only, 17 face+voice, 0 voice-only
Raw-embedding model performance	Voice [A]: EER 2.29 %, AUC 0.9967 (§3.4). Face [A]: EER 0.010, AUC 0.999 (indicative only)
Protected-template separability (128 vs 256 bit)	Documented only, raw data not committed: face AUC raw 0.935 $\rightarrow$ 256-bit 0.909 (EER 19.1 %); voice 0.997 $\rightarrow$ 0.986 (EER 5.6 %), from "20 synthetic identities $\times$ 5 samples, real checkpoints" (AUTHENTICATION_RELIABILITY_REPORT.md)

Required sentence for the paper: "Formal biometric performance metrics such as FAR, FRR, EER, and ROC-AUC cannot be established for the deployed protected-template system from the current repository."

6. Software evaluation [C]

Item	Result	Source
Backend tests	559 collected, 559 passed, 0 failed (on the exact committed content, before 873c13e)	<code>python -m pytest</code>
New tests in this update	47 (<code>test_modality_metrics.py</code> 23, <code>test_display_names.py</code> 24)	collect-only
Frontend typecheck	<code>tsc -b</code> : 0 errors	npm scripts
Frontend lint	oxlint: 0 errors, 25 warnings (in code that predates this update)	<code>npm run lint</code>
Frontend build	<code>vite build</code> : success	<code>npm run build</code>
Frontend unit tests	none exist in the project	package.json
API smoke tests	POST /users 201; empty name 422; the same name twice gives distinct IDs; CORS preflight 200	live server on a temporary DB
Browser test	Headless Chrome with synthetic camera and microphone: name validation, registration, camera stream (640×480) → backend NO_FACE (correct), voice enrollment ("Excellent", 4 sets), voice grant/deny, name shown only on grant. No real face or voice was captured by the auditor. The audit log shows 2 real face authentications under the new metric by someone else (§3.5).	smoke scripts

Largest test groups: `test_flexible_auth` 40, `test_voice_enrollment_quality` 36, `test_template_sets` 26, `test_display_names` 24, `test_modality_metrics` 23, `test_key_continuity` 18, `test_face_quality_gating` 18, `test_transform` 17, `test_voice_preprocessing` 17, `test_security_validation` 15, `test_audit_logging` 15 (54 files in total). Most integration tests use **stub embedders** (MTCNN cannot detect faces in synthetic images), so they validate the wiring, not recognition accuracy.

7. Dataset audit

Modality	Dataset	Public	Subjects	Samples	Split	License	Evidence
Face (training)	LFW via <code>sklearn.fetch_lfw_people</code> , <code>min_faces_per_person=20</code>	yes	62 identities	3,023 images	NOT AVAILABLE (val_acc 0.913 reported)	UMass non-commercial research	docs/DATASETS.md, PROJECT_REPORT.md §8.2
Voice (training / eval)	VoxCeleb1 subset (Indian celebrities), Kaggle gaurav41/...	yes (subset mirror)	NOT AVAILABLE	751 test utterances	70/15/15 (<code>training_config.json</code>)	DbCL-1.0 as declared by the mirror; not verified against VGG	docs/DATASETS.md, voice_metrics.csv
Fingerprint	SOCOFing	yes	600 (dataset)	6,000 (dataset); 900 evaluated	NOT AVAILABLE	non-commercial academic	docs/DATASETS.md
Iris	CASIA-Iris-Thousand mirror	licence-gated	—	—	—	No completed checkpoint; not part of the system	PROJECT_REPORT.md §8.4
Real users	none formally collected	—	NOT AVAILABLE	54 unlabelled attempts (local only)	—	no consent protocol documented	§3.5

Demographics: **NOT AVAILABLE** for all datasets. Augmentation: voice config lists augmentations but `augmentation_enabled: false`.

8. Mathematical formulation (only what the code implements)

Normalization (every embedder): $\hat{x} = x / \|x\|_2$

Cosine and Euclidean distance. For unit vectors, $\cos(x, y) = x \cdot y$ and $d(x, y) = \|x - y\|_2 = \sqrt{2 - 2\cos \theta}$. This identity is valid here because all embeddings are L2-normalized (`base_embedder.py`) and BioHash re-normalizes.

BioHash (§2.3): $b = \pi(1[W \hat{x} > t])$, with $W \in \mathbb{R}^{256 \times D}$, $t \sim N(0, (0.5 \cdot \text{std}(W\hat{x}))^2)$ and $(W, t, \pi) = \text{HKDF}(\text{MASTER_SECRET}, \text{context})$.

Hamming ($N = 256$):

- $H(a, b) = \sum_i 1[a_i \neq b_i]$
- $h(a, b) = 1 - H/N$ (the implemented "Hamming similarity")

Calibrated estimates [B/E]:

- $\hat{c} = h^{-1}(h)$, with $h(c) = 1 - (p_1\theta + p_2\theta^2 + p_3\theta^3)/\pi$, clamped to $[-0.2, 1]$
- $\hat{d} = \sqrt{2 - 2\hat{c}}$

Decisions:

- face: $\hat{c} \geq \tau_f = 0.80$
- voice: $\hat{d} \leq \tau_v = 0.75$
- fingerprint: $h \geq 0.90$

Fusion-scale scores (higher = better):

- $s_{\text{face}} = \hat{c}_{\text{face}}$
- $s_{\text{voice}} = 1 - \hat{d}^2/2$ ($= \hat{c}_{\text{voice}}$)
- $s_{\text{fp}} = \hat{c}_{\text{fp}}$
- thresholds: $t_{\text{face}} = 0.80$, $t_{\text{voice}} = 1 - \tau_v^2/2 = 0.71875$, $t_{\text{fp}} = \hat{c}(0.90) = 0.9384$

Fusion over the submitted set M :

- $S = (1/|M|) \sum_{m \in M} s_m$ (equal weights)
- $T = (1/|M|) \sum t_m$

Policy (`fusion/policy.py`):

- ALL_REQUIRED: $\text{accept} \Leftrightarrow \forall m: \text{match}_m$. S is reported but does not decide.
- WEIGHTED: $\text{accept} \Leftrightarrow S \geq T \wedge \forall m: s_m \geq 0.0$.
- AT_LEAST_TWO: exactly 3 modalities submitted; $\text{accept} \Leftrightarrow$ at least 2 match.

Directions: cosine $\uparrow$ better · Euclidean $\downarrow$ better · fusion $\uparrow$ better · Hamming distance $\downarrow$ better (Hamming similarity $\uparrow$ better).

9. Proposed tables

Table 1 — System configuration [D]:

Parameter	Value
Modalities	face and voice (fingerprint optional)
Face model / dimension	InceptionResnetV1 (VGGFace2 pretrained, LFW fine-tuned), 512
Voice model / dimension	ECAPA-TDNN (ArcFace-trained, VoxCeleb1 subset), 192
Normalization	L2
Template	BioHash, 256 bits (32 B), HKDF-SHA256 keyed
Template sets per user	4 (1 ACTIVE)
Face threshold	estimated cosine ≥ 0.80
Voice threshold	estimated Euclidean distance ≤ 0.75
Fusion	equal-weight mean, ALL_REQUIRED

Table 2 — Biometric metrics:

Modality	Metric	Direction	Threshold	Hamming equivalent	Runtime representation	Threshold origin
Face	cosine (estimate)	$\uparrow$	0.80	$h \geq 0.8176$ ($H \leq 46$)	$\hat{c}$ from template Hamming	teacher-requested
Voice	Euclidean (estimate)	$\downarrow$	0.75	$h \geq 0.7823$ ($H \leq 55$)	$\hat{d} = \sqrt{2 - 2\hat{c}}$	teacher-requested
Fingerprint	Hamming similarity	$\uparrow$	0.90	$H \leq 25$	measured h	inherited default

Modality	Metric	Direction	Threshold	Hamming equivalent	Runtime representation	Threshold origin
Fusion	mean of fusion-scale scores	↑	0.759375 (face+voice)	—	S	derived; not decisive under ALL_REQUIRED

Table 3 — Calibration results [B]: the §4 table.

Table 4 — Authentication results: NOT AVAILABLE for the deployed system. A real-dataset table of the embedding models (voice [A], face [A], clearly labelled "raw embedding, not protected template") is possible now.

Table 5 — Software validation [C]: the §6 table.

Table 6 — Storage / template characteristics [D/E]:

Item	Value
Per template	32 bytes plus row metadata (21 columns)
Per user, face + voice, 4 sets	$4 \times 2 \times 32 \text{ B} = 256 \text{ B}$ of template payload; revoked rows are retained with status REVOKED
Also stored	display name, internal ID, key/template/set versions, audit rows with per-modality scores, a hash fingerprint of MASTER_SECRET
Not stored	raw images, raw audio, embeddings

Table 7 — Comparison with existing approaches: qualitative only (design features: modality set, template protection used, revocability). **No numeric comparison**, because there is no comparable evaluation of the deployed system.

10. Proposed figures

#	Title	x / y	Data	Exists?	Generate now?
1	System architecture	—	§2	yes	yes
2	Face / voice pipelines	—	§2.1–2.2	yes	yes
3	BioHash generation	—	§2.3	yes	yes
4	Fusion and decision	—	§8	yes	yes
5	Calibration: embedding cosine vs Hamming similarity (means ± SD, fitted curve; face and voice)	cos / h	biohash_metric_calibration.json	yes	yes
6	Voice: Hamming similarity vs estimated Euclidean distance, with the 0.75 line	h / $\hat{d}$	same JSON + $d = \sqrt{(2-2c)}$	yes	yes
6b	Single-estimate SD vs cosine	cos / SD	JSON cosine_estimate_std	yes	yes
7	Score distributions, genuine vs impostor (deployed system)	score / density	NOT AVAILABLE	no	no
8	ROC / DET of the deployed system	FAR / TAR	NOT AVAILABLE	no	no
8a	ROC of the voice embedding model (raw)	FPR / TPR	voice_roc.csv	yes	yes, if labelled "raw embedding"
9	Template-set unlinkability histogram	h	only summary statistics in template_set_diversity.csv; per-pair data needs a rerun of evaluation.template_set_experiments	partial	after a rerun

11. Claims audit

A. Safe to claim

- The system uses 256-bit keyed BioHash cancelable templates (HKDF-SHA256 keys, orthonormal projection, keyed quantization and permutation).
- Raw face images, raw voice audio and embeddings are not stored. Only templates plus metadata are.
- Face and voice are fused under an ALL_REQUIRED policy with equal-weight score fusion.
- Templates are revocable through a pool of 4 keyed template sets. On synthetic embeddings, templates of the same embedding under different keys give $h = 0.500 \pm 0.029$ (360 pairs, 20 users). Revocation drops the revoked set's similarity to 0.504 [B].
- Display names are separate from internal identifiers.
- 559 automated backend tests pass [C].

- The voice embedding model achieves EER 2.29 % / AUC 0.9967 on a held-out VoxCeleb1-subset split, **at the raw embedding level** [A].

B. Claims that need qualification

- "Face similarity is measured by cosine similarity": say that it is a **calibrated estimate derived from the template Hamming comparison**, with $SD \approx 0.05$ near the threshold.
- "Voice uses Euclidean distance": same wording, as an **estimated** distance.
- "The calibration is accurate (R^2 0.9999)": say this is **fit quality on synthetic vectors**, justified by the transform's rotation invariance, and not validated on real embedding pairs.
- "Unlinkability / revocability": say this was **observed on synthetic embeddings**, with no formal unlinkability metric.
- "Face EER 0.010": say it is a training-log summary on LFW, the protocol is unverified, and it is raw-embedding level.
- The 128→256-bit improvement: documented, but its data is not committed.

C. Must not claim

- Any accuracy, FAR, FRR, EER, AUC or TAR **of the deployed system**.
- "Exact cosine similarity" or "exact Euclidean distance" at runtime.
- That the 0.80 / 0.75 thresholds are optimal, calibrated or validated.
- Irreversibility, "secure against attacks", "zero privacy risk", spoof or deepfake resistance.
- Any number of test users or subjects in real-user trials.
- Suitability or certification for critical infrastructure. Present it as design motivation only.
- That 559 tests represent biometric trials.

12. Limitations

1. No labelled real-user evaluation of the protected-template system; FAR/FRR/EER unknown.
2. Calibration is synthetic. The rotation-invariance argument is sound, but it has not been validated on real embedding pairs.
3. Single-attempt estimate uncertainty is about ± 0.05 in cosine at the face threshold, a 256-bit quantization effect.
4. Thresholds are teacher-requested, not tuned. The face rule ($h \geq 0.818$) is stricter than the previous 0.80 rule, and past real face scores clustered near 0.78–0.87.
5. No presentation-attack / liveness detection; spoofing and deepfake robustness not evaluated.
6. No formal security analysis (stolen-key scenario, inversion or hill-climbing attacks). BioHash security relies on the secrecy of `MASTER_SECRET`, and literature shows BioHash degrades when the token or key is compromised.
7. Voice: a 192-d embedding into 256 bits uses non-orthogonal projection blocks. Training used a limited subset (speaker count unknown).
8. Face model evaluation is closed-set LFW with an unverified protocol; fingerprint is weak (EER 30.8 %).
9. No demographic or bias analysis; no robustness study (lighting, camera, microphone, noise).
10. No latency or throughput benchmark; only incidental, cold-start-dominated audit latencies on unknown hardware.
11. Several docs are stale (§15).

13. System performance

- **Benchmarks:** none in the repository. There is no timing of the embedding, BioHash or comparison stages.
 - **Incidental end-to-end latency** from `audit_logs.latency_ms`: server-side, from request receipt to decision, including model load on cold start. Hardware is **NOT AVAILABLE**.
 - R1, 50 real attempts: median 1,808 ms, IQR 1,191–3,071 ms, mean 3,945 ms, SD 6,633 ms, p95 19,329 ms, min 316 ms, max 30,696 ms.
 - R2, 2 real face attempts: 399 ms and 1,269 ms.
 - Multi-modality requests deliberately release model caches (`authentication.release_modality_cache`, for 512 MB hosts), which inflates latency.
 - **Throughput, memory, CPU: NOT AVAILABLE.**
 - **Storage:** see Table 6 (legitimately computed).
-

14. Research contribution analysis

Implemented contributions (defensible):

1. An end-to-end, open, reproducible multimodal (face + voice) authentication system in which **only** 256-bit cancelable templates are stored.
2. **Metric-specific decision rules on protected templates.** Face cosine and voice Euclidean decisions are expressed through an offline-calibrated mapping from template Hamming similarity, and fused on one direction-consistent scale. This is **an engineering method, and it needs literature positioning**: SimHash/random-hyperplane angle estimation is classical, so claim the *application and calibration within a keyed BioHash*, not the idea itself.
3. Revocation of whole multimodal credentials through keyed template-set pools.
4. A usability-oriented registration design that separates display name from identifier.

Novelty claims to avoid unless a literature search supports them: "first", "novel fusion", "novel BioHash". Suggested wording: "we implement and evaluate...", "we adapt...", "to our knowledge, few open implementations..." (only if supported by the search).

15. Documentation vs implementation discrepancies

#	Document	Says	Implementation / data says
1	docs/PROJECT_REPORT.md §8.2	fingerprint EER 0.445, AUC 0.579	fingerprint_metrics.csv: EER 0.3077, AUC 0.7644 (later checkpoint, commit 3a0bfeb)
2	docs/PROJECT_REPORT.md	24 tests; Phase 2/3 "not yet implemented"	559 tests; template protection, fusion and UI implemented
3	README "The problem"	face, iris , fingerprint	the application uses face, voice and fingerprint; iris has no completed checkpoint
4	Real-user face range	0.738–0.820 (deleted face_threshold.json note)	the same report elsewhere says 0.738–0.859 over 15 attempts; the source log is not in the repo
5	PROJECT_DOCUMENTATION.md, VIVA_TEMPLATE_PROTECTION_PREP.md	Hamming 0.80 thresholds	superseded (banner added in 873c13e)
6	template_protection/biohash.py	DEFAULT_OUTPUT_BITS = 128	runtime uses 256 (Settings.template_bits); cosmetic, but state 256 in the paper
8	scripts/calibrate_biohash_metric_mapping.py::MODALITY_DIMS, evaluation/template_set_experiments.py docstring	fingerprint 256-d	the fingerprint embedder is 512-d. Only fingerprint's fusion-scale score is affected (its decision is $h \geq 0.90$); the D=512 face curve is nearly identical. Fix: set 512, rerun the calibration (~15 s), rerun tests
7	docs/BIOMETRIC_METRICS.md	"about ± 0.05 near the thresholds"	0.050 at $c = 0.80$; 0.057–0.063 at $c \approx 0.72$ (voice threshold); still accurate as "about"

16. Recommended paper structure (IEEE)

1. Abstract
2. Introduction (motivation, unimodal limits, template-protection need, contributions)
3. Related work (§17 categories)
4. Threat model and design goals (privacy of stored templates, revocability; out of scope: liveness)
5. System architecture
6. Face and voice pipelines
7. Cancelable protection (BioHash, HKDF, template sets)
8. Metric-specific decisions on protected templates and calibration
9. Fusion and decision policy
10. Implementation and user workflow (display names)
11. Experimental setup: A calibration, B embedding-model evaluation, C template-set experiments, D software testing, E real-user evaluation (only if done)
12. Results
13. Discussion

14. Security and privacy considerations
15. Limitations
16. Conclusion and future work
17. References

Abstract ingredients (verified status in brackets)

- **Problem:** stored biometric templates are irrevocable [context].
- **System:** face + voice, 256-bit BioHash, HKDF keys [D].
- **Decisions and fusion:** estimated cosine ≥ 0.80 and estimated distance ≤ 0.75 , fused under ALL_REQUIRED [D].
- **Calibration:** $R^2 \geq 0.9999$ on 19,600 synthetic pairs per modality, single-estimate SD ≈ 0.05 [B].
- **Voice model:** EER 2.29 % at the raw-embedding level [A].
- **Unlinkability:** cross-key similarity 0.500 on synthetic embeddings [B].
- **Software:** 559 tests [C].
- **Limitation:** no deployed-system FAR/FRR [statement].

17. Recommended references

These are cited from knowledge. **Verify every entry (DOI or URL) before submission.** None were fetched during this audit.

Category	Reference	Why
Biometrics foundations	A. K. Jain, A. Ross, S. Prabhakar, "An introduction to biometric recognition," IEEE TCSVT 14(1):4–20, 2004, doi:10.1109/TCSVT.2003.818349	framing
Multimodal / fusion	A. Ross, A. K. Jain, "Information fusion in biometrics," Pattern Recognition Letters 24(13):2115–2125, 2003, doi:10.1016/S0167-8655(03)00079-5	fusion levels
Fusion	A. Jain, K. Nandakumar, A. Ross, "Score normalization in multimodal biometric systems," Pattern Recognition 38(12):2270–2285, 2005, doi:10.1016/j.patcog.2005.01.012	score normalization and sum rule
Fusion	J. Kittler et al., "On combining classifiers," IEEE TPAMI 20(3):226–239, 1998, doi:10.1109/34.667881	sum rule
Fusion	K. Nandakumar et al., "Likelihood ratio-based biometric score fusion," IEEE TPAMI 30(2):342–347, 2008, doi:10.1109/TPAMI.2007.70796	alternative fusion (future work)
Multibiometrics	A. Ross, K. Nandakumar, A. K. Jain, <i>Handbook of Multibiometrics</i> , Springer, 2006, doi:10.1007/0-387-33123-9	background
Cancelable biometrics	N. K. Ratha, J. H. Connell, R. M. Bolle, "Enhancing security and privacy in biometrics-based authentication systems," IBM Systems Journal 40(3):614–634, 2001, doi:10.1147/sj.403.0614	origin of cancelable biometrics
Cancelable biometrics	N. K. Ratha et al., "Generating cancelable fingerprint templates," IEEE TPAMI 29(4):561–572, 2007, doi:10.1109/TPAMI.2007.1004	transforms
Cancelable review	V. M. Patel, N. K. Ratha, R. Chellappa, "Cancelable biometrics: A review," IEEE Signal Processing Magazine 32(5):54–65, 2015, doi:10.1109/MSP.2015.2434151	survey
Template protection	A. K. Jain, K. Nandakumar, A. Nagar, "Biometric template security," EURASIP J. Adv. Signal Process. 2008:579416, doi:10.1155/2008/579416	requirements (revocability, non-invertibility)
Template protection	K. Nandakumar, A. K. Jain, "Biometric template protection: Bridging the performance gap between theory and practice," IEEE SPM 32(5):88–100, 2015, doi:10.1109/MSP.2015.2427849	performance/security trade-off
Template protection survey	C. Rathgeb, A. Uhl, "A survey on biometric cryptosystems and cancelable biometrics," EURASIP J. Info. Security 2011:3, doi:10.1186/1687-417X-2011-3	survey
BioHash	A. T. B. Jin, D. N. C. Ling, A. Goh, "Biohashing: two factor authentication featuring fingerprint data and tokenised random numbers," Pattern Recognition 37(11):2245–2255, 2004, doi:10.1016/j.patcog.2004.04.011	original BioHash
BioHash analysis	A. Kong et al., "An analysis of BioHashing and its variants," Pattern Recognition 39(7):1359–1368, 2006, doi:10.1016/j.patcog.2005.10.025	stolen-token weakness (limitation)
BioHash	A. B. J. Teoh, Y. W. Kuan, S. Lee, "Cancellable biometrics and annotations on BioHash," Pattern Recognition 41(6):2034–2044, 2008, doi:10.1016/j.patcog.2007.12.002	BioHash properties
BioHash	A. Lumini, L. Nanni, "An improved BioHashing for human authentication," Pattern Recognition 40(3):1057–1065, 2007, doi:10.1016/j.patcog.2006.05.030	variants
Unlinkability	M. Gomez-Barrero et al., "General framework to evaluate unlinkability in biometric template protection systems," IEEE TIFS 13(6):1406–1420, 2018, doi:10.1109/TIFS.2017.2788000	metric for future work
Binary hashing / angle	M. S. Charikar, "Similarity estimation techniques from rounding algorithms," STOC 2002, doi:10.1145/509907.509965	$P(\text{bit differs}) = \theta/\pi$ (basis of the calibration)

Category	Reference	Why
Random hyperplane	M. X. Goemans, D. P. Williamson, JACM 42(6):1115–1145, 1995, doi:10.1145/227683.227684	same principle
Face detection	K. Zhang et al., "Joint face detection and alignment using multitask cascaded convolutional networks," IEEE SPL 23(10):1499–1503, 2016, doi:10.1109/LSP.2016.2603342	MTCNN
Face embedding	F. Schroff, D. Kalenichenko, J. Philbin, "FaceNet," CVPR 2015, doi:10.1109/CVPR.2015.7298682	embedding and L2 metric
Face backbone	C. Szegedy et al., "Inception-v4, Inception-ResNet...", AAAI 2017, doi:10.1609/aaai.v31i1.11231	InceptionResNet
Face training data	Q. Cao et al., "VGGFace2," IEEE FG 2018, doi:10.1109/FG.2018.00020	pretraining
Face loss	J. Deng et al., "ArcFace," CVPR 2019, doi:10.1109/CVPR.2019.00482	loss used
Face dataset	G. B. Huang et al., "Labeled Faces in the Wild," UMass TR 07-49, 2007, http://vis-www.cs.umass.edu/lfw/	dataset
Speaker embedding	B. Desplanques, J. Thienpondt, K. Demuynck, "ECAPA-TDNN," Interspeech 2020, doi:10.21437/Interspeech.2020-2650	voice model
Speaker embedding	D. Snyder et al., "X-vectors," ICASSP 2018, doi:10.1109/ICASSP.2018.8461375	background
Speaker dataset	A. Nagrani, J. S. Chung, A. Zisserman, "VoxCeleb," Interspeech 2017, doi:10.21437/Interspeech.2017-950	dataset
Toolkit	M. Ravanelli et al., "SpeechBrain," arXiv:2106.04624, 2021	implementation
Spoofing (voice)	M. Todisco et al., "ASVspoof 2019," Interspeech 2019, doi:10.21437/Interspeech.2019-2249	limitation / future
Spoofing (general)	S. Marcel et al. (eds.), <i>Handbook of Biometric Anti-Spoofing</i> , 2nd ed., Springer, 2019, doi:10.1007/978-3-319-92627-8	PAD
Standards	ISO/IEC 24745:2022 (biometric information protection); ISO/IEC 30107-3 (PAD testing); ISO/IEC 19795-1 (performance testing)	evaluation protocol
Key derivation	H. Krawczyk, P. Eronen, RFC 5869 (HKDF), 2010, https://www.rfc-editor.org/rfc/rfc5869	key derivation
Identity guidelines	NIST SP 800-63B, doi:10.6028/NIST.SP.800-63b	biometric authenticator requirements
Critical infrastructure	NIST SP 800-82 Rev. 3, 2023, doi:10.6028/NIST.SP.800-82r3	context only
Bias	P. Grother, M. Ngan, K. Hanaoka, NISTIR 8280, 2019, doi:10.6028/NIST.IR.8280	demographic-effects limitation
Fingerprint dataset	Y. I. Shehu et al., "SOCOFing," arXiv:1807.10609, 2018	dataset

18. Missing experiments / data and paper-readiness assessment

Readiness: a system/implementation paper can be written now if it stays within §11 A–B. It is **not ready** for claims of authentication performance. Priority 1–2 below would move it to an evaluation paper.

WHAT WE STILL NEED TO MEASURE

1. Real-user genuine/impostor evaluation of the deployed protected-template system (highest priority).

- **Why:** it is the only way to report FAR/FRR/EER/ROC of what was built.
- **Data:** consented participants (document consent and data deletion). Each person enrolls once: 5 face poses and 2 voice recordings. Then ≥ 10 genuine face and ≥ 10 genuine voice attempts across 2 sessions on different days. Recommended ≥ 30 people (minimum 20).
- **Protocol:**
 - Genuine: each person's own attempts against their own template.
 - Impostor: each person's attempts against every other person's template **under the target's key**, which is what the real system does (`scripts/calibrate_protected_thresholds.py` documents this pairing).
 - Record the Hamming similarity per comparison and never store raw embeddings beyond the offline research harness.
 - Volume: $30 \times 10 = 300$ genuine and $30 \times 10 \times 29 = 8,700$ impostor comparisons per modality.
- **Metrics:** FAR, FRR, TAR at 0.80/0.75 and at EER; EER; ROC-AUC; the same for face-only, voice-only and fusion (ALL_REQUIRED and WEIGHTED); bootstrap 95 % CIs.
- **Graphs:** genuine/impostor histograms (Hamming and estimated metric), ROC, DET, FAR/FRR vs threshold.
- **Tables:** a per-modality and per-fusion results table.

- **Files:** `evaluation/real_user_evaluation.py` → `evaluation/results/real_user_scores.csv` (anonymized IDs, modality, genuine flag, h , $\hat{c}/\hat{d}$) and `real_user_metrics.csv`.

2. Real-embedding validation of the calibration (can be done now, with no participants).

- **Why:** it confirms the synthetic rotation-invariance assumption on real embeddings.
- **Data:** LFW (face, 512-d, via the repo checkpoint) and the VoxCeleb subset (voice, 192-d): all pairs, or $\geq 20,000$ sampled pairs, with the true cosine computed offline.
- **Protocol:** template both embeddings under the same random key and compare the true cosine with $\hat{c}$ from `metric_estimation`.
- **Metrics:** RMSE, MAE, bias and SD of $(\hat{c} - c)$ binned by c ; the same for $\hat{d}$.
- **Graphs:** $\hat{c}$ vs c scatter with the identity line; error vs c .
- **Files:** `scripts/validate_calibration_real.py` → `evaluation/results/calibration_real_validation.csv`.

3. Protected-template vs raw-embedding separability on public datasets (can be done now).

- **Why:** it quantifies the accuracy cost of template protection.
- **Data:** LFW and the VoxCeleb subset test split.
- **Metrics:** EER and AUC for raw cosine vs 256-bit h (same key for genuine, target key for impostors).
- **Graphs:** ROC overlay.
- **Files:** `evaluation/results/{face,voice}_protected_vs_raw.csv`. This also puts the currently doc-only 128-vs-256 table on committed data.

4. Threshold sensitivity.

- **What:** FAR/FRR as the face threshold varies over 0.70–0.90 and the voice threshold over 0.55–1.0 (from #1 or #3).
- **Output:** curves plus a table at the teacher values. Save to `evaluation/results/threshold_sensitivity.csv`.

5. Latency benchmark.

- **Protocol:** 100 warm and 10 cold runs per stage (preprocess, embed, BioHash, compare, full request), recording the hardware (CPU, RAM, OS) and single vs multi-modality requests.
- **Metrics:** mean, median, SD, p95.
- **Script:** `scripts/benchmark_latency.py` → `evaluation/results/latency_benchmark.csv`.

6. Unlinkability and revocability with real embeddings.

- **Protocol:** Gomez-Barrero $D \leftrightarrow / D_{\text{sys}}$ on mated vs non-mated cross-key template pairs from #3. Revoked-vs-new similarity distribution.
- **Output:** `evaluation/results/unlinkability_real.csv` plus a histogram.

7. Presentation-attack testing (ISO/IEC 30107-3).

- **Attacks:** printed photo, screen replay and video for face; replayed recording and TTS/voice-clone for voice; ≥ 10 attacks per species per participant subset.
- **Metrics:** APCER, BPCER.
- **Output:** `evaluation/results/pad_results.csv`. Expect poor results, since no liveness check exists; report honestly.

8. Stolen-key scenario.

- **Protocol:** an attacker holding `MASTER_SECRET` or the user's key: impostor Hamming distribution when the attacker uses the target key (the #1 impostor protocol already models this), plus the known BioHash stolen-token analyses.
- **Output:** a discussion plus the #1 numbers.

9. Robustness.

- **Face:** lighting levels and camera distance.
- **Voice:** SNR 20/10/5 dB noise and two microphones.
- **Metric:** genuine-score shift and FRR at the thresholds.
- **Output:** `evaluation/results/robustness.csv`.

10. Demographic analysis.

- Only with explicit consent and a sufficient sample; otherwise state it as a limitation.

Also: commit the face evaluation's pair-level data (ROC CSV), the split protocol and the speaker count for voice. Keep the data-deletion and consent record with the study.